

LESSON 8-4 • SESSION 1 • GUIDED NOTES

Writing Inequalities

Learning target – I can write an inequality that describes a constraint in a real situation. · 6.EE.B.8

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
inequality <i>desigualdad</i>	A math sentence that _____ using $<$, $>$, $\leq$ or $\geq$.	$x \geq 18$
at least <i>al menos</i>	That number _____ ($\geq$).	at least 5: $x \geq 5$
at most <i>a lo sumo</i>	That number or _____ ($\leq$).	at most 8: $x \leq 8$
more than <i>más que</i>	_____ than the number, not equal to it ($>$).	more than 5: $x > 5$

KEY IDEA*complete each sentence*

- First decide whether the boundary number is _____.
- Allowed: use $\leq$ or _____. Not allowed: use $<$ or _____.

WORKED EXAMPLE*fill in the blanks as we solve*To make the team, a swimmer must swim more than 20 laps. Write an inequality for L , the laps.**Step 1** Key words: "_____."**Step 2** Is 20 allowed? ____, so use $>$.**Step 3** Inequality: L ____ 20.**Answer** $L > 20$, so 20 laps is ____ a solution.**YOU TRY***use the example as a model***1.** Write an inequality for each phrase. Use x .**a.** at least 13 ____**b.** fewer than 9 ____**2.** A coupon works on orders of \$50 or less. Write an inequality for d , the order amount. Does a \$50 order work?*Show your work.***Answer** _____**SUM IT UP***in your own words*

To choose an inequality symbol, I ____ because ____.

LESSON 8-4 • SESSION 2 • GUIDED NOTES

Representing Inequalities

Learning target – I can represent an inequality with a set of solutions and describe them. · 6.EE.B.8

WORDS TO KNOW*fill in the missing words*

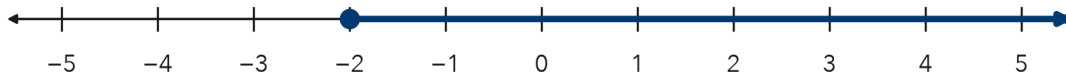
WORD	WHAT IT MEANS	EXAMPLE
open circle <i>círculo abierto</i>	The boundary is ____ a solution ($<$ or $>$).	$x > 3$: open circle at 3
closed circle <i>círculo cerrado</i>	The boundary ____ a solution ($\leq$ or $\geq$).	$x \leq 3$: closed circle at 3
solutions of an inequality <i>soluciones de una desigualdad</i>	____ the numbers that make it true.	$x > 3$: 3.5, 4, 10, ...

KEY IDEA*complete each sentence*

- Draw a circle at the _____ number.
- For $<$ or $\leq$, the arrow points _____. For $>$ or $\geq$, it points _____.

WORKED EXAMPLE*fill in the blanks as we solve*Graph $x < 3$.**Step 1** The boundary is ____.**Step 2** $<$ does not include 3, so draw an ____ circle.**Step 3** Draw the arrow to the ____.**Answer** Sample solutions: 2, 0, -1.5**YOU TRY***use the example as a model*

1. Write the inequality shown. Use
- x
- .



Inequality _____

2. Is each number a solution of
- $n > 7$
- ? Write yes or no.

a. 7 ____

b. 7.5 ____

SUM IT UP*in your own words*

To graph an inequality, I ____ because ____.
